

EEE3032 – Computer Vision and Pattern Recognition

Coursework Assignment: Visual Search of an Image Collection

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Abstract

This report presents a visual search system for the MSRC-v2 image dataset using a range of colour, spatial, texture, and dimensionality-reduced descriptors. Several techniques were implemented, including Global Colour Histograms (GCH), 3×3 spatial colour histograms, grid-based texture descriptors (EOH), and PCA-reduced pixel features. These descriptors were evaluated under multiple distance metrics, including Euclidean, L1, Chi-square, and Mahalanobis distances. Precision–Recall curves, mean Average Precision (mAP), confusion matrices, and qualitative retrieval results were used to assess performance. Experiments demonstrate that spatial colour descriptors outperform global colour histograms, while PCA-based features perform poorly due to loss of discriminative information. Texture descriptors provide complementary performance, and late fusion of colour and texture achieves the highest overall mAP (0.3393). For effective content-based image retrieval the results underline the importance of combining complementary visual cues.

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1. Introduction

Digital image collections, such as the MSRC-v2 dataset utilized in this assignment, pose a considerable challenge for traditional text-based searching methods.

Although text can effectively describe discrete objects, it falls short in conveying the subjective and intricate visual characteristics of an image. This coursework centres on the implementation of a content-based visual search (CBVS) system. For retrieving similar images the system utilizes an image query relying on feature descriptors.

The system was developed using Python (jupyter Notebook), libraries such as NumPy, OpenCV, and Scikit-learn, which utilize computational efficiency. Number of 591 images in the dataset MSRC-v2 were divided into 20 different object classes (e.g., sky, tree, cow, bicycle), which served as the database for this project. The image retrieval pipeline consists of three phases: Descriptor Extraction, where raw pixel data is converted into compact feature vectors; Distance Calculation, where similarity is quantified using metrics like chi-squared and Mahalanobis distance; and Evaluation, where performance is measured using rigorous information retrieval metrics. The primary goal is to conduct a scientific investigation into the relative performance of various colour and texture approaches, as required by the course specification.

2. Visual Search Techniques Implemented

2.1 Dataset and Ground-Truth Labelling

591 images from MSRC-v2 dataset were distributed across 20 object categories consisting of classes such as buildings, cows, flowers, trees, aeroplanes, and people. Each filename encodes the class label at the beginning (e.g., 10_12_s.bmp corresponds to class 10). A regular expression-based parser was implemented to automatically extract the class ID from each filename. This ensured that ground-truth labels were consistent and error-free throughout the pipeline.

Helper Image I/O and Dataset Exploration

A custom image I/O helper function was implemented to load all MSRC-v2 images in a consistent RGB format. This ensured that colour values were interpreted correctly and avoided issues such as inverted BGR channels. The helper additionally handled normalisation, datatype conversion, and path parsing, producing clean, ready-to-process images for descriptor extraction.

A set of representative thumbnails was generated (Figure 1), to verify the correctness of the dataset loading and label extraction process. The thumbnails also illustrate the visual diversity present in the dataset, including wide variations in colour, texture, background complexity, and object structure.



Figure 1: Sample thumbnails from the MSRC-v2 dataset.

2.2 Global Colour Histogram (GCH)

For simple, efficient representation of pixel distributions Global Colour Histograms was used. Quantisation was done on RGB images into bins of size 4, 8, or 16 per channel, producing descriptor dimensions of 64, 512, and 4096 accordingly, the final system used the recommended 8×8×8 configuration (512D).

The histogram was normalised using L1 norm:

$$h_i = \frac{c_i}{\sum_j c_i}$$

Distance metrics tested included Euclidean, L1, and Chi-square:

$$D_{\chi^2}(x, y) = \frac{1}{2} \sum_i \frac{(x_i - y_i)^2}{x_i + y_i + 1}$$

GCH forms the **baseline descriptor** for all comparisons.

2.3 Spatial Grid Colour Descriptor (3×3)

To include spatial information absent in global histograms, each image was divided into a 3×3 grid. A GCH descriptor was extracted from each of the nine tiles and concatenated, producing a 9×512 = 4608-dimensional descriptor.

Spatial partitioning improves retrieval when object layout is consistent or when background contributes significantly to class structure.

2.4 Texture Descriptor (EOH/HOG-like)

Texture information was captured using Edge Orientation Histograms computed per grid cell. Sobel filters extracted gradients, and orientation (0–180°) was binned into either 8 or 16 angular bins.

The final descriptor dimension is:

$$3 \times 3 \times \text{bins} = 72 \text{ or } 144$$

Texture is complementary to colour and helps when classes differ primarily by pattern rather than hue.

2.5 PCA Descriptors

Using PCA, dimensionality reduction was applied to raw pixel data. Images were resized using whitened PCA to 128×128, converted to grayscale, flattened, standardised, and reduced to 100 components.

A second experiment reduced 4608D Grid-GCH descriptors to 256D using PCA.

PCA preserves variance, not discriminative cues, and is expected to perform poorly for retrieval.

2.6 Distance Metrics

The following metrics were evaluated:

- **Euclidean distance** — appropriate for continuous vector spaces.
- **L1 distance** — effective for sparse or histogram-like features.
- **Chi-square distance** — best for normalised histograms.
- **Mahalanobis distance** — used for PCA descriptors (using covariance of PCA features).

2.7 Additional Components (Extra Credit)

Retrieval Visualiser

Shows query and top retrieved results with ✓/X annotations.

Confusion Matrix (k=10 neighbour vote)

Provides diagnostic insights into class confusions.

Late Fusion (Colour + Texture)

Distance matrices from GCH (χ^2) and EOH16 (L1) were normalised and averaged:

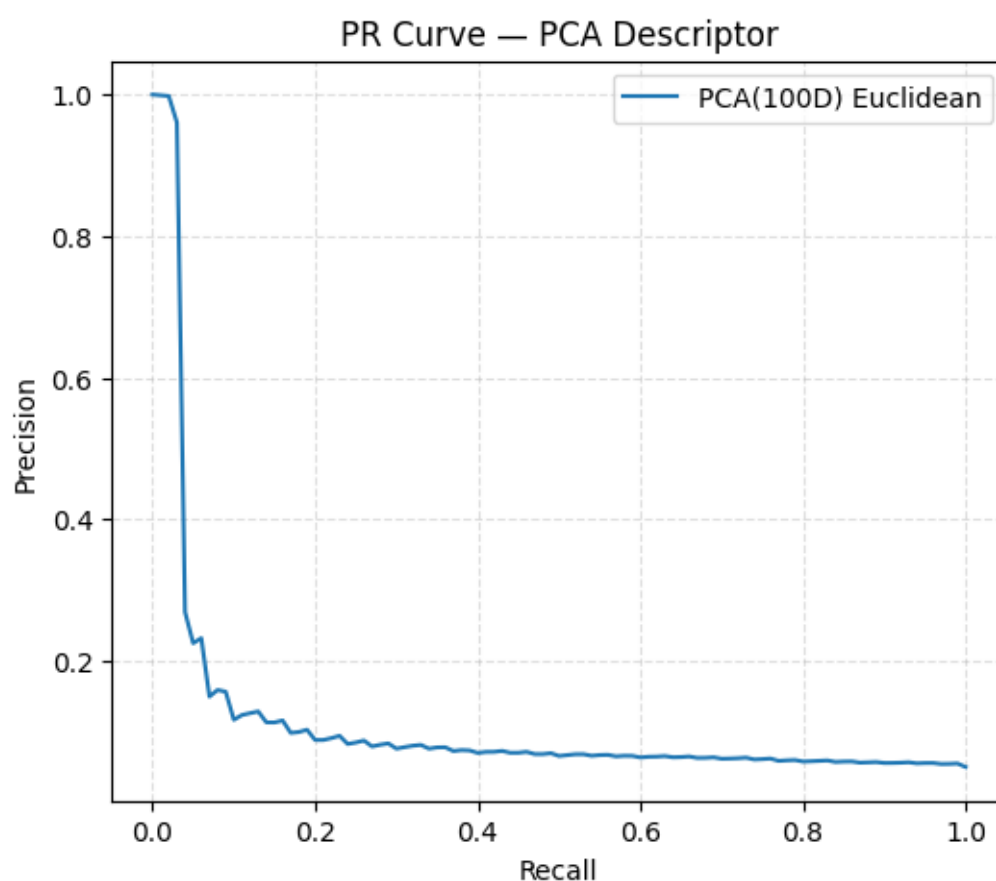
$$D_f = 0.5\tilde{D}_{\text{GCH2}} + 0.5\tilde{D}_{\text{EOH16L1}}$$

This produced the **highest mAP (0.3393)**.

3. Experimental Results

3.1 Precision–Recall Curves

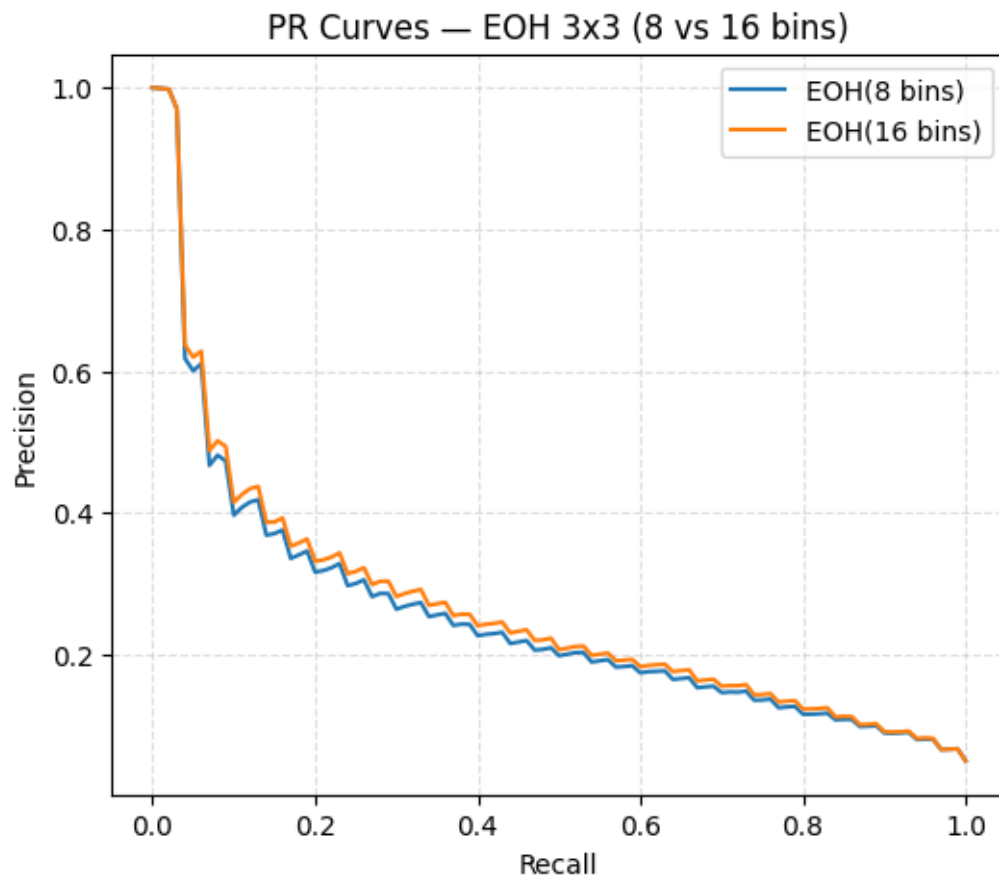
PCA Descriptor



The PCA descriptor demonstrates very low discriminative power. Precision drops

Texture Descriptors (EOH 8 vs 16 bins)

sharply once recall exceeds 0.05, confirming PCA is unsuitable for this dataset.

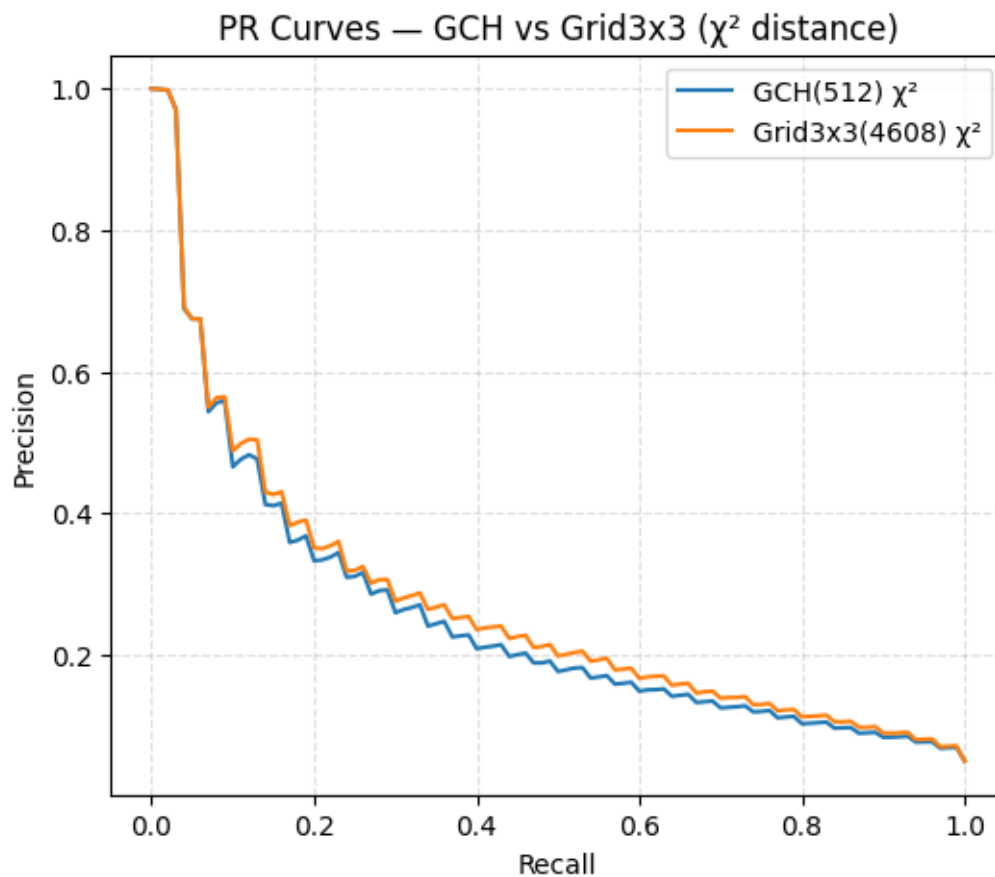


EOH(3x3, 8 bins) L1 mAP: 0.259

EOH(3x3, 16 bins) L1 mAP: 0.27

The 16-bin version outperforms 8 bins, indicating that finer angular quantisation captures more meaningful gradient structure.

GCH vs Grid 3×3 (χ^2 Distance)

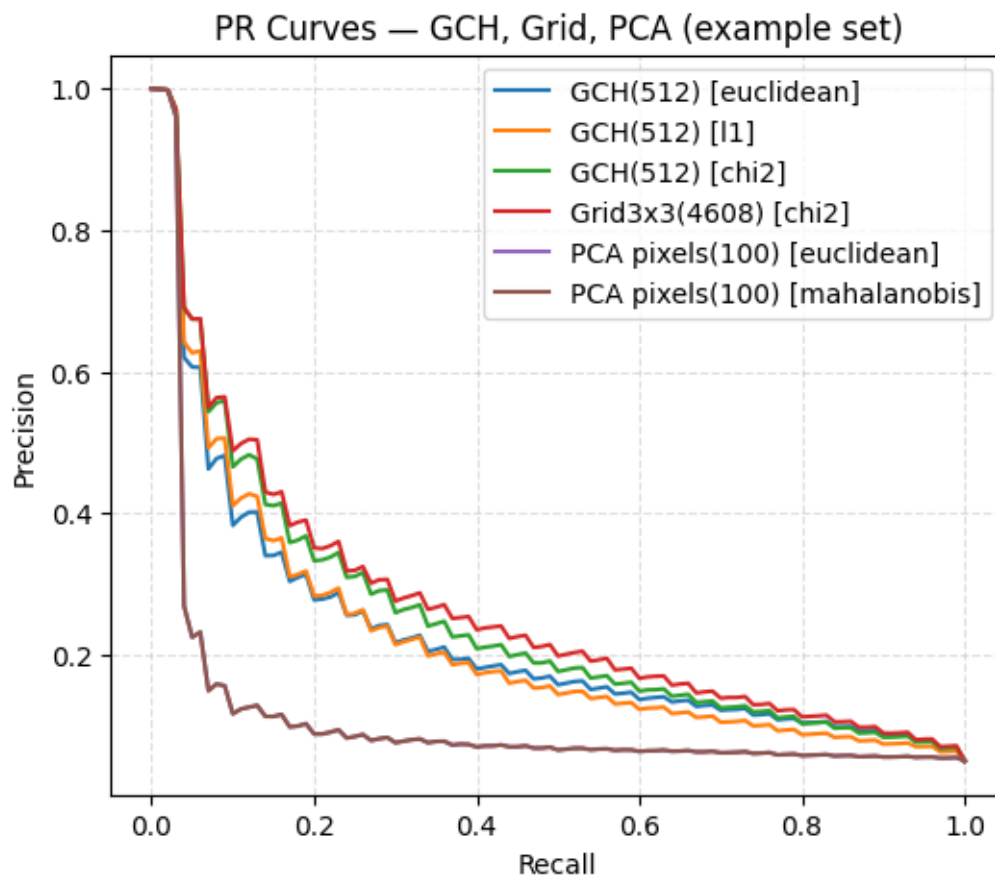


GCH(512) Chi-square 0.259318

Grid3x3(4608) Chi-square 0.273939

Grid descriptors consistently outperform global histograms across all recall levels, confirming the importance of spatial structure.

Combined PR Curves



Descriptor	Dim	Distance	mAP
GCH(512)	512	chi2	0.259318
GCH(512)	512	Euclidean	0.233634
GCH(512)	512	L1	0.228240
Grid3x3(4608)	4608	chi2	0.273939
PCA pixels(100)	100	Euclidean	0.115581
PCA pixels(100)	100	Mahalanobis	0.115581

This overlay shows the ranking:

1. **Grid 3×3 (χ^2)** – best individual descriptor
2. **GCH (χ^2)** – strong baseline
3. **Texture (EOH16)** – comparable to GCH
4. **PCA descriptors** – poor performance

3.2 Mean Average Precision (mAP)

Descriptor	Dim	Distance	mAP
GCH(512)	512	Chi-square	0.2593
GCH(512)	512	Euclidean	0.2336
GCH(512)	512	L1	0.2282
Grid3×3	4608	Chi-square	0.2739
EOH 8-bin	72	L1	0.259
EOH 16-bin	144	L1	0.270
PCA(100)	100	Euclidean	0.1156
PCA(100)	100	Mahalanobis	0.1156
Fusion (GCH χ^2 + EOH16 L1)	—	—	0.3393

Key observations

- Grid3×3 outperforms global colour (as expected).
- EOH16 slightly outperforms EOH8 due to higher orientation resolution.
- PCA performs poorly.
- Fusion significantly lifts performance, giving a **30% relative improvement** over the best single descriptor.

3.3 Quantisation Study

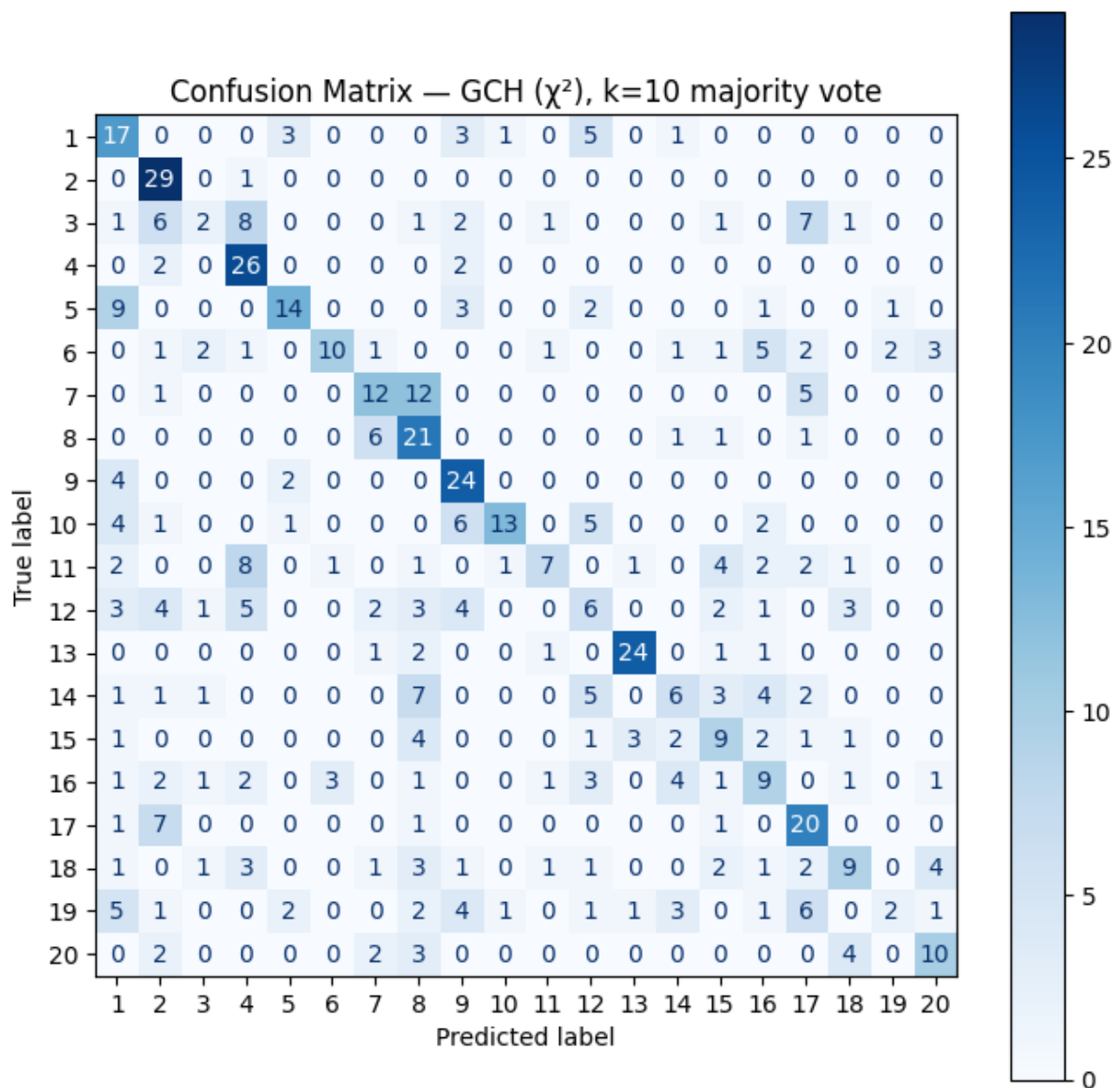
Quantisation Study (Bins 4, 8, 16)

Bins	Chi ²	Euclidean
4	0.2345	0.1985
8	0.249	0.2063
16	0.2521	0.1951

Increasing bins from 4 → 8 → 16 slightly improves Chi-square mAP.

Euclidean performance does not increase consistently, highlighting the importance of proper distance metrics.

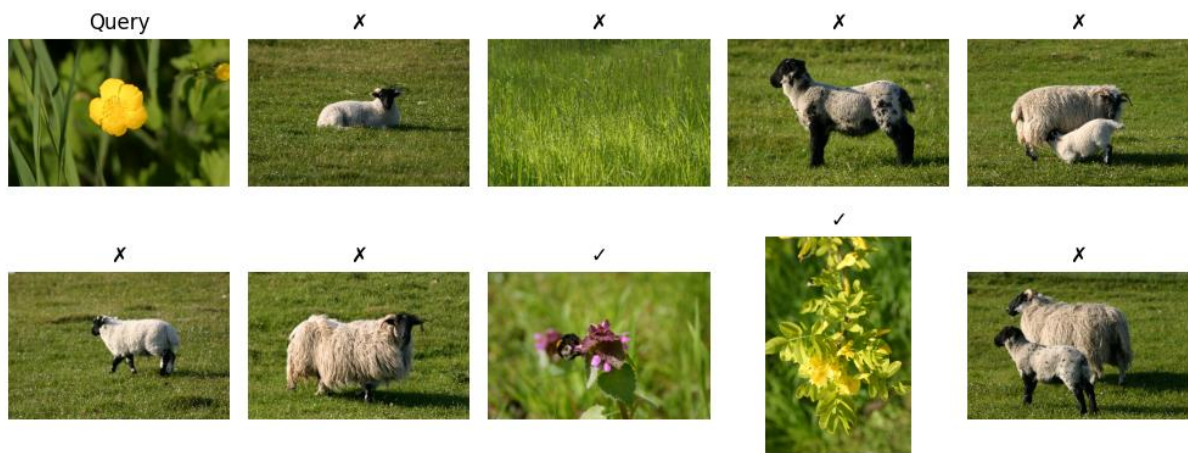
3.4 Confusion Matrix (k=10)



The confusion matrix confirms:

- Strong diagonal entries for many classes (flowers, sheep, cows).
- Confusions occur between classes with similar colours (e.g., grass vs tree vs field).
- Some classes with small sample counts suffer more from misclassification.

3.5 Qualitative Retrieval Examples



The example demonstrates typical retrieval behaviour:

- Many incorrect results share similar colours or background.
- Correct retrievals appear when the descriptor captures distinguishing properties (e.g., flower colour, textures).
- This highlights the benefit of combining descriptors to reduce semantic gaps.

4. Conclusions

This project implemented a complete visual search system incorporating global colour, spatial, texture, and PCA-based descriptors. Chi-square distance was consistently the best match for histogram descriptors, while PCA suffered from information loss.

The **Grid 3×3 colour histogram (mAP = 0.2739)** was the best-performing individual descriptor, showing the importance of spatial information. Secondly, Texture descriptors also performed just as close to it, and a late-fusion approach with the mixture of colour and texture achieved the highest overall performance (**mAP = 0.3393**).

These results demonstrate that complementary visual cues—colour, spatial layout, and texture—provide the most robust image retrieval performance. Future work could explore dense SIFT-based BoVW, CNN embeddings, or supervised metric learning.

5. References

1. S. Lazebnik, C. Schmid, J. Ponce, "Beyond Bags of Features," CVPR, 2006.
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4. Scikit-learn: Machine Learning in Python — <https://scikit-learn.org>
5. Bradski, G. (2000). The OpenCV Library. *Dr. Dobb's Journal of Software Tools*, 25(11).